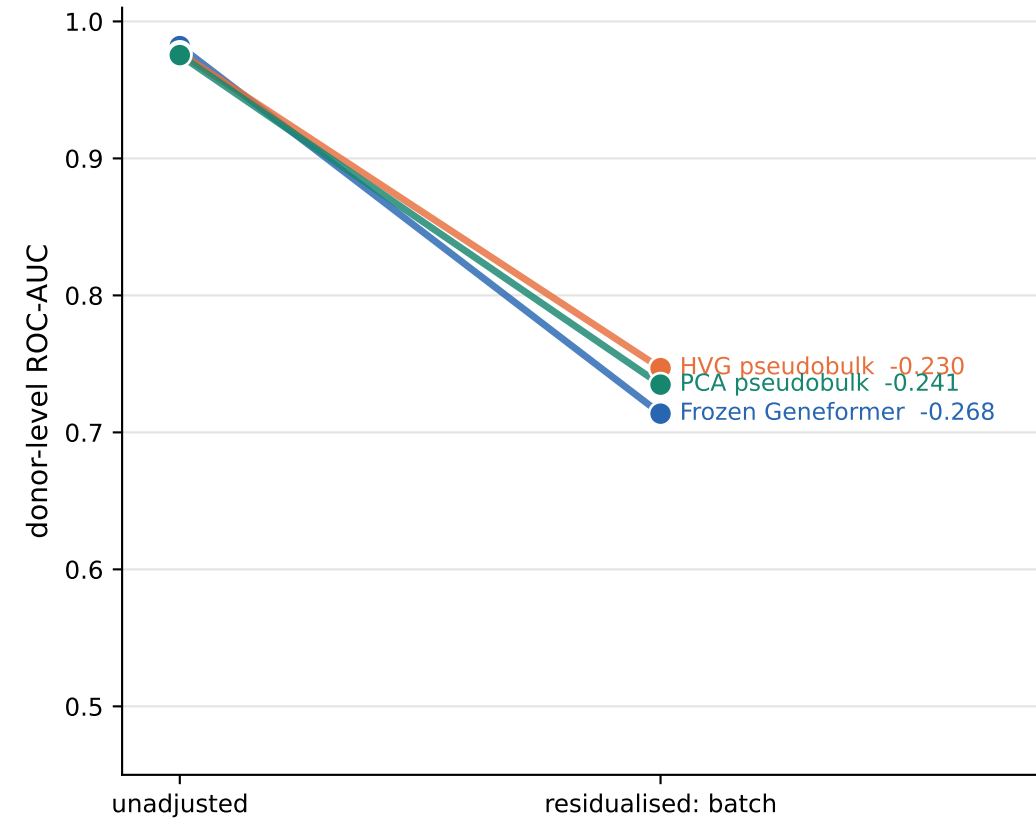


Figure 6 Residualisation measures sensitivity, not a causal batch effect

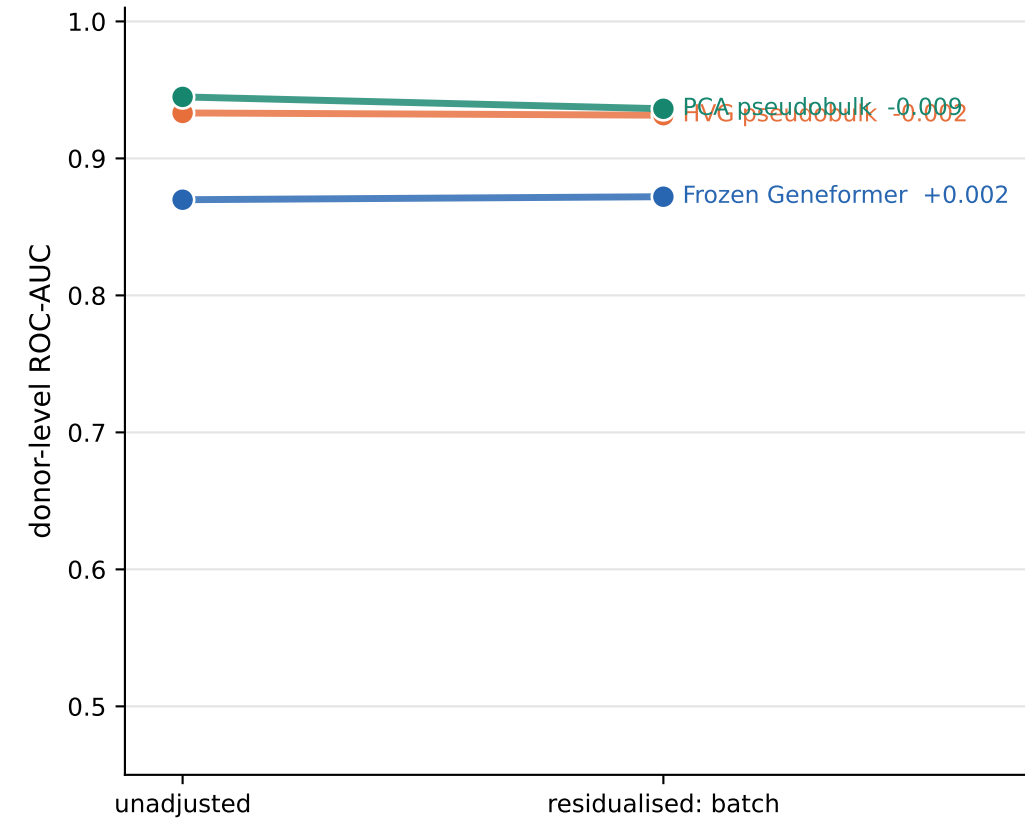
A Wave residualisation removes 0.23-0.27 AUC

wave-only disease AUC = 0.922



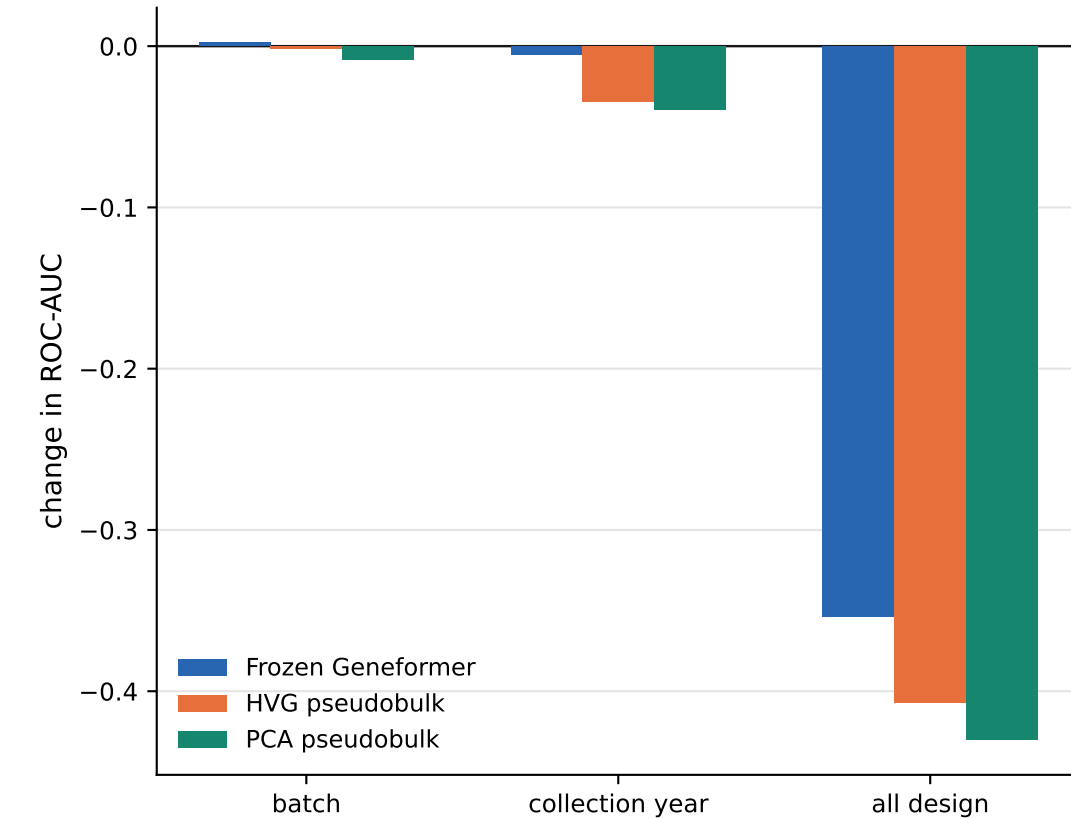
B Batch exposure is not batch dependence

best representation-to-batch AUC = 0.993; batch-only disease AUC = 0.499



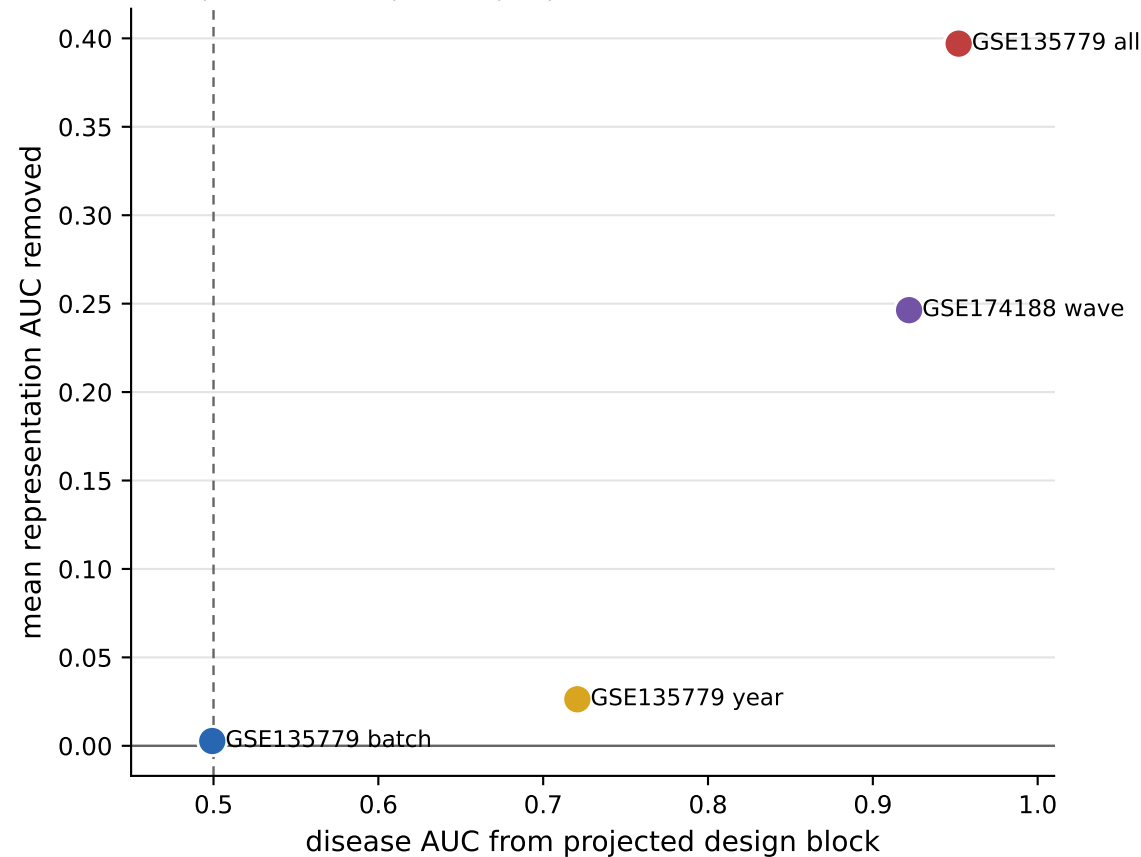
C Damage follows the projected design block

identical folds and classifier pipeline



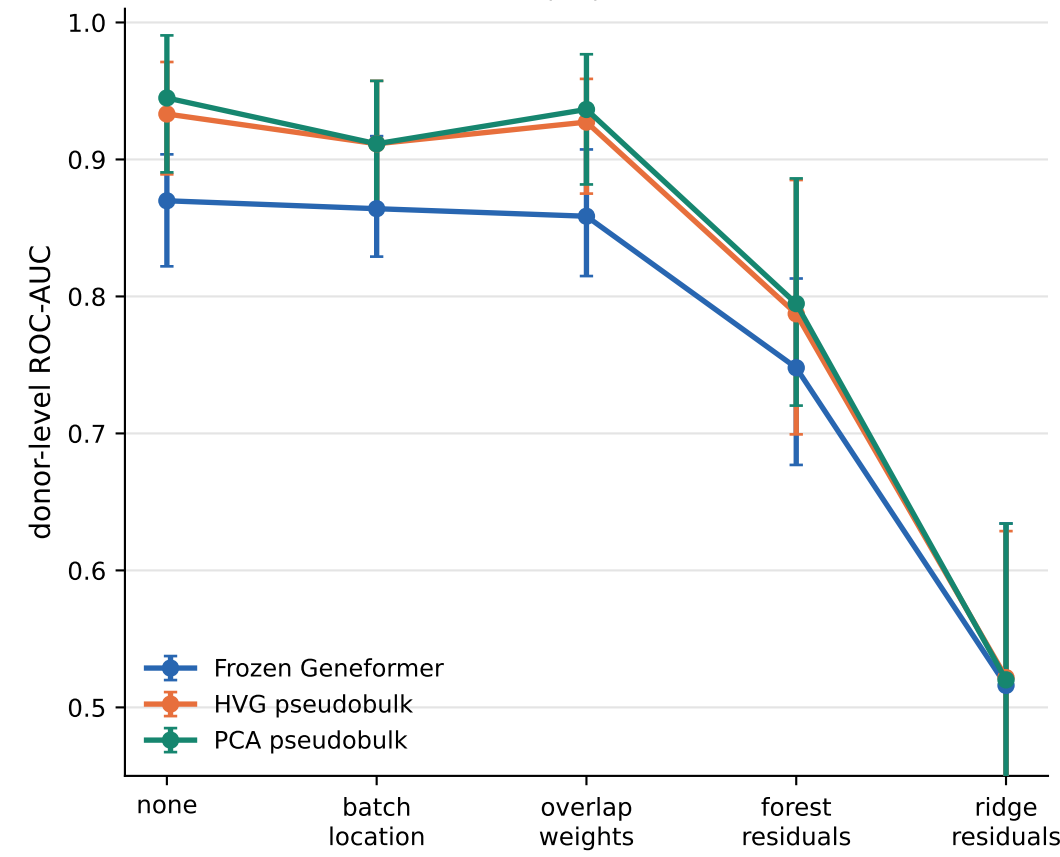
D Design-label alignment forecasts attenuation

descriptive relationship across prespecified blocks



E The conclusion survives nonlinear adjustment

GSE135779; bars are 2.5th-97.5th split percentiles



F The attenuation gap remains positive

paired repeated-split bootstrap; descriptive split-sensitivity interval

